

8) Week 14: Initial value problems

8.1) Consider the following ordinary differential equation:

$$y'' + \alpha y'y + y = t$$

- With $\alpha = 1$, is the ODE *linear* or *nonlinear*?
- With $\alpha = 0$, is the ODE *linear* or *nonlinear*?
- With $\alpha = 1$, what is the order of the ODE?
- With $\alpha = 0$, what is the order of the ODE?
- Express the ODE in *vector form* (keeping α as a variable).
- Use Euler's method to estimate the value of $y(1)$, starting from $y(0) = 2$, $y'(0) = 1$, and with a step size of $h = 1$. Express the solution as a function of α .

a) $\alpha=1$: $y'' + \underbrace{y'y}_{\text{nonlinearity}} + y = t$... nonlinear

b) $\alpha=0$ $y'' + y = t$... linear.

c) 2nd order

d) 2nd order

e) $y_1 = y$, $y_2 = y'$, $\gamma = (y_1, y_2)$

$$\begin{aligned} \therefore \gamma' &= (y'_1, y'_2) = (y_2, y'') \\ &= (y_2, -\alpha y'_1 y_1 - y_1 + t) \\ &= (y_2, -\alpha y_2 y_1 - y_1 + t) = f(\gamma, t). \end{aligned}$$

or equivalently:

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ -\alpha y_2 y_1 - y_1 + t \end{bmatrix}$$

$$f) \quad \hat{Y}[0] = \begin{bmatrix} y'(0) \\ -\alpha y'(0) y(0) - y(0) + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -\alpha \cdot 1 \cdot 2 - 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2\alpha - 2 \end{bmatrix}$$

$$\hat{Y}[1] = \hat{Y}[0] + f(\hat{Y}[0], 0) \cdot h$$

$$= \begin{bmatrix} \hat{y}_1[0] \\ \hat{y}_2[0] \end{bmatrix} + f\left(\begin{bmatrix} \hat{y}_1[0] \\ \hat{y}_2[0] \end{bmatrix}, 0\right) \cdot (1)$$

$$= \begin{bmatrix} \hat{y}_1[0] \\ \hat{y}_2[0] \end{bmatrix} + \begin{bmatrix} \hat{y}_2[0] \\ -\alpha \hat{y}_2[0] \hat{y}_1[0] - \hat{y}_1[0] + 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -\alpha(1)(2) - 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2\alpha - 1 \end{bmatrix}$$

$$\therefore \hat{y}(1) = 3$$

8.2) Rewrite the following second-order linear ODE into a system of first-order ODEs.

$$y'' - y' + y = 0$$

- a) Is this a linear ODE or a non-linear ODE?
- b) Write the ODE in matrix form.
- c) With initial condition $y(0) = 1$, $y'(0) = 1$ and step size $h = 1$, perform one step of Euler's method and estimate $y(1)$.
- d) Perform one step of the midpoint method.
- e) Perform one step of Heun's method.

a) linear

b) $\gamma = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ with $y_1 = y$, $y_2 = y'$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ y_2 - y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\therefore \gamma' = A\gamma \text{ with } A = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}.$$

c) Euler:

$$\hat{\gamma}_{[0]} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \hat{\gamma}_{[1]} &= \hat{\gamma}_{[0]} + h f(\hat{\gamma}_{[0]}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

d) Midpoint

$$S_0 = f(\hat{\gamma}_{[0]}) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\gamma_m = \hat{\gamma}_{[0]} + \frac{h}{2} S_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$$

$$S_m = f(y_m) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \hat{S}$$

$$\hat{y}_{[1]} = \hat{y}_{[0]} + h \hat{S} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

e) Heun.

$$S_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$y_1 = \hat{y}_{[0]} + h S_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$S_1 = f(y_1) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\hat{S} = \frac{1}{2} (S_0 + S_1) = \frac{1}{2} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}.$$

$$\hat{y}_{[1]} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

8.3) Consider the following ODE:

$$y''' + 3t^2 y' - 2y = e^{-t}$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first iteration of the Heun's method with initial condition $y = 0$, $y' = 1$, $y'' = 0$, and time step $h = 1$.

a) Third order.

b) Linear

c) $y_1 = y$ $Y = (y_1, y_2, y_3)$
 $y_2 = y'$
 $y_3 = y''$

$$\begin{aligned} Y' = (y_1', y_2', y_3') &= (y_2, y_3, e^{-t} - 3t^2 y' + 2y) \\ &= (y_2, y_3, e^{-t} - 3t^2 y_2 + 2y_1) \end{aligned}$$

Matrix form:

$$\underbrace{\begin{bmatrix} y_1' \\ y_2' \\ y_3' \end{bmatrix}}_{Y'} = \begin{bmatrix} y_2 \\ y_3 \\ e^{-t} - 3t^2 y_2 + 2y_1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -3t^2 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_Y + \underbrace{\begin{bmatrix} 0 \\ 0 \\ e^{-t} \end{bmatrix}}_w$$

$$f(Y, t) = A(t) Y + w(t)$$

$$d) \quad \hat{y}[0] = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} s_0 &= f(\hat{y}[0], 0) = A(0) \hat{y}[0] + W(0) \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

$$y_1 = \hat{y}[0] + h s_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} s_1 &= f(y_1, h) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ e^{-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/e \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 + 1/e \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \hat{s} &= \frac{1}{2} (s_0 + s_1) = \frac{1}{2} \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -1 + 1/e \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 \\ 1/2 \\ 1/2e \end{bmatrix} \end{aligned}$$

$$\hat{y}[1] = \hat{y}[0] + h \hat{s} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1/2 \\ 1/2e \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \\ 1 + 1/2e \end{bmatrix}$$

8.4) The Duffing equation is used to model certain damped and driven oscillators. The Duffing equation is given by:

$$\beta y^3 + y'' + \delta y' + \alpha y = \gamma \cos(\omega t)$$

where δ is the damping coefficient, α and β are the linear and non-linear stiffness coefficients, respectively, and $\gamma \cos(\omega t)$ represents the driving force with amplitude γ and frequency ω .

- What is its order?
- Is it linear or nonlinear?
- Express the ODE in vector form. If linear, express it in matrix form.

a) Second order

b) Nonlinear.

$$\begin{aligned} \text{c) } \begin{matrix} y_1 = y \\ y_2 = y' \end{matrix} &\longrightarrow \begin{matrix} y_1' = y_2 \\ y_2' = y'' = \gamma \cos \omega t - \alpha y - \delta y' - \beta y^3 \\ &= \gamma \cos \omega t - \alpha y_1 - \delta y_2 - \beta y_1^3 \end{matrix} \end{aligned}$$

$$Y = (y_1, y_2)$$

$$Y' = (y_1', y_2') = (y_2, \gamma \cos \omega t - \alpha y_1 - \delta y_2 - \beta y_1^3)$$

8.5) Consider the following ODE:

$$y'' - \mu(1 - y^2)y' + y = 0$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first iteration of the midpoint method with initial condition $y = 0$, $y' = 1$, time step $h = 0.5$, and $\mu = 1$.

a) 2nd order.

b) non linear.

c) $y_1 = y$
 $y_2 = y'$ $\gamma = (y_1, y_2)$

$$\gamma' = (y_1', y_2') = (y_2, \mu(1 - y_1^2)y_2 - y_1) = f(\gamma)$$

d) $f((y_1, y_2)) = (y_2, (1 - y_1^2)y_2 - y_1)$

$$\hat{\gamma}[0] = (0, 1)$$

$$S_0 = f(\hat{\gamma}[0]) = f((0, 1)) = (1, (1 - 0)1 - 0) \\ = (1, 1)$$

$$\gamma_m = \hat{\gamma}[0] + h S_0 = (0, 1) + \frac{0.5}{2} (1, 1) \\ = (0.25, 1.25)$$

$$\begin{aligned}
 S_m = f(y_m) &= f((0.25, 1.25)) \\
 &= (1.25, (1 - 0.25^2)(1.25) - 0.25) \\
 &= (1.25, 0.921875) = \hat{S}
 \end{aligned}$$

$$\begin{aligned}
 \hat{y}[1] &= \hat{y}[0] + 0.5 \hat{S} \\
 &= (0, 1) + 0.5 (1.25, 0.921875) \\
 &= (0.625, 1.461)
 \end{aligned}$$

8.6) Consider the following ODE:

$$y'' - 4y' + 4y = 10$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first two iterations of Euler's method with initial condition $y = 0$, $y' = 0$ and time step $h = 0.1$.

a) 2nd order

b) linear

c)
$$\begin{aligned} y_1 &= y \\ y_2 &= y' \end{aligned} \rightarrow Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$Y' = \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ 10 + 4y_2 - 4y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix} Y + \begin{bmatrix} 0 \\ 10 \end{bmatrix}$$

d)
$$\hat{Y}[0] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \hat{Y}[1] &= \hat{Y}[0] + h f(\hat{Y}[0]) \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 0.1 \left(\begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} . \end{aligned}$$

$$\begin{aligned} \hat{Y}[2] &= \hat{Y}[1] + h f(\hat{Y}[1]) \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.1 \left(\begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.1 \left(\begin{bmatrix} 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 1.4 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 2.4 \end{bmatrix} \end{aligned}$$